

Experimental Fixed-Window QK Relational Architecture

Linear ordinary reading with separate strict multi-hop propagation

Companion scaling note to *Experimental Full-Pair QK Relational Architecture*

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Overview

This companion note reports a fixed-window revision of the ordinary reading path in the previously published full-pair QK Relational Architecture. The original model used a headless, no-learned-value QK field for ordinary language reading and a separate strict relational field for optional multi-hop propagation. In the original implementation both relationship fields were full-pair and therefore quadratic in sequence length.

The change studied here is deliberately narrow. Ordinary reading is restricted to a fixed causal relationship window

$$W = cd, \tag{1}$$

where d is model width and c is a fixed multiplier. The experiments reported here use the simplest tested rule, $c = 1$, so $W = d$. For a fixed model this changes ordinary relationship work from $O(N^2d)$ to $O(NWd)$, which is linear in sequence length N . The distance prior, headless QK geometry, raw-state transmission, residual/MLP structure and separate strict hop geometry are otherwise retained.

This does *not* make every possible mode of the architecture linear. In the default ordinary mode, $k = 0$, the hop projections are dormant and the model uses the cheap fixed-window path only. If explicit deeper hopping is requested, the present A_h field is still full-history and dense, so hop construction and repeated propagation remain quadratic in N . The architecture therefore has two compute regimes: linear ordinary reading and optional quadratic relational depth.

Two experiments are reported. First, $W = d = 48$ is inserted into the exact small benchmark from the earlier paper, including the answer-only hop task. Second, the language comparison is scaled to $d = 96$, $W = 96$, context 512 and 271,296 matched parameters against a genuine four-head Transformer. The scaled QK model finishes ahead on held-out loss and accuracy for all three seeds and is faster in the measured CPU timing tests, with the speed advantage increasing at longer timing contexts. Peak process RSS is not lower in the current implementation, so no measured total-memory advantage is claimed.

1 Motivation and scope

The research question for this companion note is:

Can the ordinary A_0 path be changed from full-pair reading to a fixed $W = d$ relationship window, preserving the useful language and hop behaviour while obtaining genuinely linear ordinary relationship scaling in sequence length?

A second question concerns system structure. The model already separates ordinary reading from explicit relational propagation. If ordinary reading is local and inexpensive, deeper full-history relational compute can remain optional rather than being paid on every ordinary forward pass. This is a different claim from saying that Transformers cannot use sliding windows. They can. The point tested here is whether the QK architecture’s separated ordinary and hop geometries allow its ordinary path to be made aggressively local without destroying the behaviour established in the earlier note.

The evidence is still experimental. It is one byte-level corpus, three seeds, small models by modern standards, CPU timings, and a controlled synthetic hop task. The results justify a fixed-window

QK variant as a serious candidate architecture; they do not establish general superiority over Transformers or a fully linear long-range reasoning system.

2 Fixed-window architecture

2.1 Window rule $W = cd$

Let the normalized hidden-state matrix in a block be

$$H \in \mathbb{R}^{N \times d}. \quad (2)$$

The general fixed-window rule is

$$W = cd, \quad (3)$$

with fixed c for a model family. The experiments in this note test

$$\boxed{W = d}. \quad (4)$$

Thus the $d = 48$ model uses $W = 48$, and the scaled $d = 96$ model uses $W = 96$. The rule is tied to model width rather than sequence length. If N grows while the trained model width remains fixed, W does not grow with N .

The distinction matters. A rule such as $W = N/2$ is still quadratic because the window expands with sequence length. A fixed $W = cd$ gives ordinary relationship cost proportional to N for a fixed model. If both N and d are treated as variables, the tested $W = d$ rule gives $O(Nd^2)$ ordinary relationship work, not linear scaling in model width.

2.2 Local distance-aware ordinary QK reading

The ordinary projections are unchanged:

$$Q_0 = HW_{Q_0}, \quad K_0 = HW_{K_0}. \quad (5)$$

The fixed long-tailed distance prior is also unchanged:

$$D(r) = \frac{0.02}{1+r} + \frac{0.98}{(1+r)^2}. \quad (6)$$

For query position i , define the legal ordinary key set

$$\mathcal{J}_W(i) = \{j : j \leq i, i - j < W\}. \quad (7)$$

The local ordinary weights are

$$A_{0,ij}^{(W)} = \frac{\exp\left(q_i k_j^T / \sqrt{d} + \log D(i - j)\right)}{\sum_{\ell \in \mathcal{J}_W(i)} \exp\left(q_i k_\ell^T / \sqrt{d} + \log D(i - \ell)\right)} \quad \text{for } j \in \mathcal{J}_W(i), \quad (8)$$

with zero weight outside the window. In matrix notation this is equivalently

$$A_0^{(W)} = \text{softmax}\left(\frac{Q_0 K_0^T}{\sqrt{d}} + \log D(|i - j|) + M_W\right), \quad (9)$$

where M_W permits only $0 \leq i - j < W$.

The notation is conceptual: a linear implementation does not need to construct an $N \times N$ matrix containing zeros outside the window. It can evaluate only the legal local scores and immediately apply them to the transmitted state. There is still no learned V projection and no attention-head split in the QK operator.

For ordinary operation,

$$Z_0 = A_0^{(W)} H. \quad (10)$$

The relationship part of this computation uses at most NW QK scores rather than N^2 scores.

2.3 Separate strict full-history hop field

The explicit hop mechanism is intentionally unchanged from the prior architecture. It has its own learned geometry,

$$Q_h = HW_{Q_h}, \quad K_h = HW_{K_h}, \quad (11)$$

and a strict causal no-self mask $M_<$:

$$A_h = \text{softmax} \left(\frac{Q_h K_h^T}{\sqrt{d}} + M_< \right). \quad (12)$$

For k requested extra hops,

$$\boxed{Z_k = A_0^{(W)} A_h^k H.} \quad (13)$$

The separation is important. The earlier experiments found that separate Q_h, K_h plus the strict no-self mask were both needed for reliable learned transitions. The fixed-window change is therefore applied only to ordinary A_0 reading; the hop geometry is not merged with the local field.

At $k = 0$, Q_h, K_h and A_h are not evaluated. At $k > 0$, A_h is computed once and reused for repeated multiplication, but the current full-history $Q_h K_h^T$ and each dense A_h propagation are quadratic in sequence length.

2.4 Two compute regimes

For the default ordinary mode,

$$\text{ordinary relationship work} \sim O(NWd), \quad (14)$$

$$W = cd \implies O(Ncd^2), \quad (15)$$

which is linear in N for fixed c and d .

For explicit hopping, the current dense implementation additionally pays approximately

$$O(N^2 d) \quad \text{to build } A_h, \quad (16)$$

and

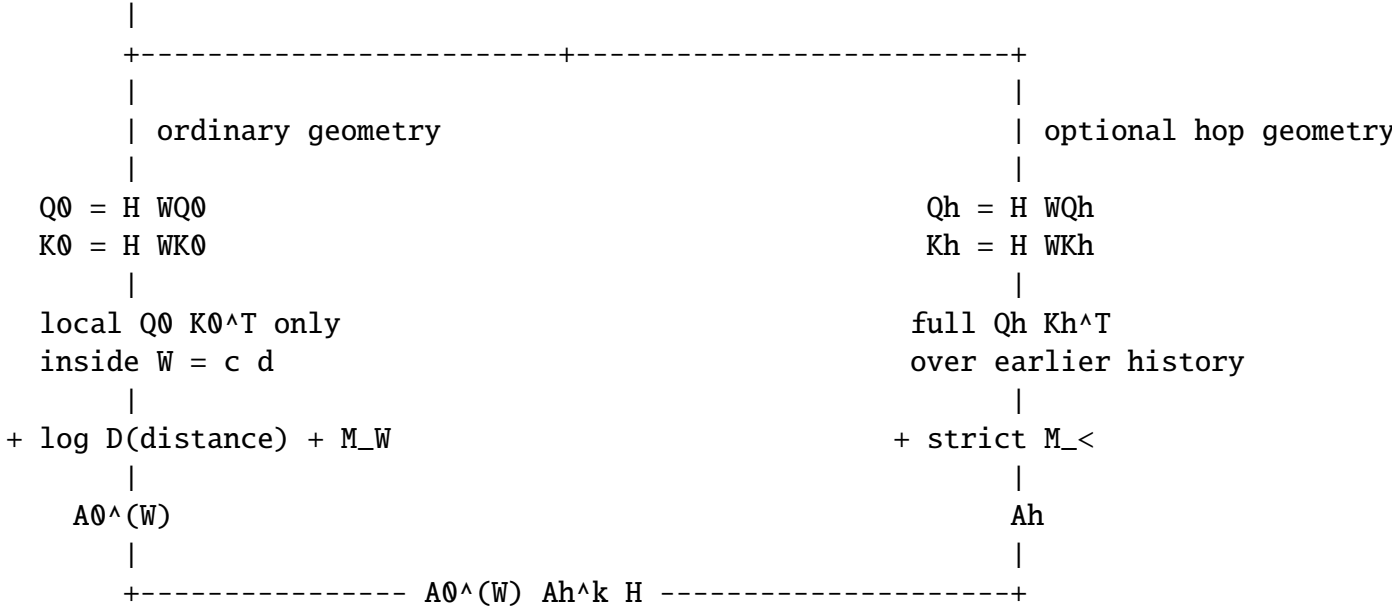
$$O(kN^2 d) \quad (17)$$

for k dense hop propagations. The hop mode is therefore still quadratic in sequence length. The design should be described as *linear ordinary reading plus optional quadratic relational depth*, not as a fully linear architecture.

If an application performs most ordinary predictions with $k = 0$ and invokes hops selectively, most ordinary forward passes avoid the quadratic hop path. The present experiments do not implement or evaluate an automatic learned policy deciding when to request hops; hop count remains an external control.

2.5 ASCII architecture sketch

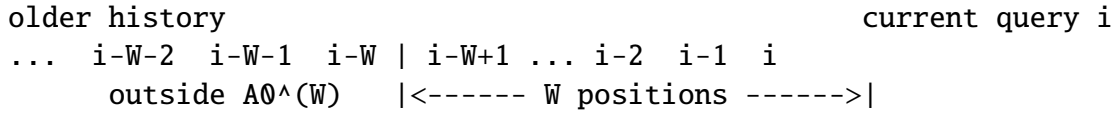
normalized token state H



$k = 0$: hop branch dormant; ordinary path is fixed-window / linear in N

$k > 0$: Ah is activated; current hop construction/propagation is quadratic

A positional view of the ordinary window is:



Ah , when requested, may still relate i to positions in the older history.

3 Implementation and verification

3.1 Exact blocked local operator

The scaled implementation uses contiguous local QK blocks. For each query block it exposes only the nearby key span required by W , applies the exact causal-window mask and the same

distance prior, computes the normalized local weights, applies them to the transmitted states, and discards the temporary score block. It never constructs a global $N \times N$ ordinary score matrix.

A schematic implementation is:

```
for each query block [s:e]:
    key_start = max(0, s - W + 1)
    q = Q0[:, s:e]
    k = K0[:, key_start:e]
    v = state_to_read[:, key_start:e]    # raw H, or Ah^k H

    score = q @ k.T / sqrt(d)
    score += log_D(distance)
    score.mask_outside(0 <= i-j < W)
    out[:, s:e] = softmax(score) @ v
```

The symbol v in this pseudocode denotes the state being transmitted through the relation weights; it is not a learned value projection.

3.2 Implementation verification

The $d = 96$, $W = 96$ scaled implementation was checked against an equivalent dense $W = 96$ masked reference. The maximum absolute discrepancy was 4.77×10^{-7} for the optimized blocked implementation. A causal-prefix check produced zero change to earlier outputs when later tokens were replaced. Both compared language models contain exactly 271,296 stored parameters, and the final-block hop parameters receive no gradient in ordinary $k = 0$ language training.

At the smaller $d = 48$, $W = 48$ scale, a separate true local implementation was also compared with the dense-window version used for rapid training; full-model logit differences were on the order of 10^{-5} , consistent with floating-point reordering. The W48 hop benchmark retained the strict no-self A_h rule and the same characteristic depth-33 terminal/readout failure seen in the earlier full-pair paper, rather than making the known negative result disappear.

4 Experimental protocol

4.1 Direct bridge to the earlier benchmark: $d = 48$, $W = 48$

The first test changed only the ordinary relationship visibility rule in the earlier language benchmark. It used the same Vim 9.1 `version9.txt` byte stream, the same split boundaries, byte vocabulary 256, context 128, batch 8, 4,096 optimizer updates, AdamW at 5×10^{-4} , weight decay 0.01, gradient clipping 1.0, and seeds 11, 22 and 33. Each seed/model saw 4,194,304 next-byte training targets and was evaluated on the same 32,768 held-out targets.

The model remained two layers and width $d = 48$ with exactly 80,352 stored parameters. The only language-architecture change was ordinary $A_0 \rightarrow A_0^{(48)}$. A $W = 128$ parity control reproduced the original seed-11 early training trace exactly, and the held-out test-data hash matched the earlier benchmark.

The answer-only hop protocol was also rerun with $W = 48$: 33 chain edges plus 15 distractor edges, fresh graph mappings, no edge-transition labels, no hop-depth feature in the input, 500 normal-only steps followed by 1,000 mixed depth-1 through depth-8 steps, and final-answer cross-entropy only. The sequence length is 49 including the query, so $W = 48$ genuinely removes one oldest ordinary key at the final query while A_h remains full-history.

4.2 Scaled language benchmark: $d = 96, W = 96$

The second test increased model width and training context while retaining a strict matched comparison. Both models use two layers and exactly 271,296 trainable parameters. QKRA uses width 96, $W = 96$, first-block feed-forward width 480 and final-block feed-forward width 384, with separate stored Q_h, K_h only in the final block. The Transformer uses four heads of 24 dimensions each and feed-forward width 384.

Training uses context 512, batch 4, 4,096 optimizer updates, AdamW at 5×10^{-4} , weight decay 0.01, gradient clipping 1.0 and seeds 11, 22 and 33. Each seed/model therefore sees

$$4096 \times 4 \times 512 = 8,388,608 \quad (18)$$

next-byte target presentations, twice the target count of the earlier small benchmark per seed/model. Evaluation scores 65,536 fixed held-out targets per seed.

The runs are checkpointed every 128 optimizer updates, including model state, optimizer state, RNG state, step count and training history. This makes the experiment resumable without restarting completed work or changing the data-sampling sequence.

5 Results

5.1 $W = d = 48$ bridge result

Seed	W48 loss	Full-QK loss	T loss	W48 acc.	Full-QK acc.	T acc.
11	1.7323	1.7324	1.7510	52.85%	52.51%	51.75%
22	1.7538	1.7789	1.7476	52.68%	52.39%	51.76%
33	1.7358	1.7434	1.7800	52.59%	52.74%	50.50%
Mean	1.7406	1.7516	1.7595	52.71%	52.55%	51.34%

The mean W48-minus-Transformer loss difference is -0.0189 nats/byte; the paired three-seed 95% t-interval is $[-0.0815, +0.0437]$ and crosses zero. The mean W48-minus-full-QK difference is -0.0109 nats/byte with interval $[-0.0430, +0.0211]$. The defensible conclusion is therefore not that W48 is superior, but that no quality penalty was detected at this scale from restricting ordinary reading to $W = d = 48$.

5.2 Hop compatibility at $W = 48$

Seed	Depth 2	Depth 8	Depth 16	Depth 24	A_h top-1	Correct mass
11	76.9%	75.6%	75.7%	74.1%	99.994%	99.679%
22	39.6%	31.5%	33.2%	28.1%	100.000%	99.481%
33	72.9%	71.1%	72.5%	70.1%	99.988%	99.663%

The same seed pattern remains: seeds 11 and 33 retain strong deep-hop end-answer behaviour, seed 22 has weaker end-answer convergence, while the learned transition field is essentially perfect for all three seeds. Depth 33 remains a terminal/readout failure and is not counted as solved. The fixed ordinary window therefore did not erase the behaviour that motivated retaining a separate strict A_h field.

5.3 Scaled $d = 96$, $W = 96$ competitor result

Seed	QKRA-W96 loss	Transformer loss	QKRA acc.	Transformer acc.
11	1.6033	1.7044	56.52%	52.86%
22	1.6093	1.7745	56.61%	50.83%
33	1.6068	1.7375	56.14%	51.98%
Mean	1.6065	1.7388	56.42%	51.89%

The mean paired QKRA-minus-Transformer loss difference is -0.1323 nats/byte. The paired 95% t-interval over the three seeds is $[-0.2119, -0.0527]$. The mean accuracy difference is $+4.53$ percentage points with a three-seed paired interval of $[1.78, 7.28]$ percentage points. Both intervals exclude zero in this experiment, but $n = 3$ remains too small for a broad claim of architectural superiority.

The stronger conclusion is narrower: the $W = d$ local ordinary path did not merely survive the original small test. At a model with more than three times the parameter count and four times the training context of the earlier paper, it remained competitive and finished ahead of the matched full-attention four-head Transformer on all three tested seeds.

6 Runtime, arithmetic and memory

6.1 Saved-run training time

The complete checkpointed 4,096-update training runs used the explicit exact blocked local-QK implementation. Mean elapsed training time was:

Model	Total time	Mean ms/update
QKRA-W96	156.34 s	38.17 ms
4-head Transformer	197.60 s	48.24 ms

The Transformer/QKRA elapsed ratio was 1.264x in this CPU setup. These are implementation- and hardware-specific measurements, not a universal speed ratio.

6.2 Isolated millisecond and peak-RSS measurements

For isolated timing, the QK path used the mathematically identical $W = 96$ blocked implementation backed by an optimized attention primitive inside each local block. The primitive is used only as a numerical kernel: there is still no learned value projection and no key outside the $W = 96$ span is exposed to the local operation.

Mode	Context / batch	QKRA ms	T ms	T/QK	QKRA RSS	T RSS
infer	512 / 4	13.78	15.94	1.16x	275.1 MiB	273.3 MiB
train	512 / 4	32.83	45.22	1.38x	368.6 MiB	364.2 MiB
infer	1024 / 1	4.88	9.23	1.89x	269.9 MiB	267.1 MiB
infer	2048 / 1	15.07	25.59	1.70x	277.9 MiB	272.9 MiB
infer	4096 / 1	28.19	58.47	2.07x	295.9 MiB	285.8 MiB
infer	8192 / 1	55.60	172.01	3.09x	336.2 MiB	320.9 MiB
train	2048 / 1	49.33	82.63	1.68x	368.5 MiB	364.1 MiB
train	4096 / 1	83.10	204.10	2.46x	415.2 MiB	413.1 MiB

The long-context rows are runtime scaling probes, not evidence of long-context language quality. The model was trained at context 512, and robust quality extrapolation beyond that context has not been established.

The speed direction is nevertheless consistent with the complexity change. At context 512/batch 4, inference is only 1.16x faster. At context 8,192/batch 1, the measured inference ratio grows to 3.09x. The whole network cannot inherit the full theoretical pair-count ratio because feed-forward layers, projections, normalization, output logits, memory movement and framework overhead remain.

6.3 Major-MAC and relationship geometry

Using the same approximate accounting convention as the earlier competitor audit and excluding costs common to both models, context 512/batch 4 gives approximately 490.7M counted major MACs for QKRA-W96 versus 855.6M for the Transformer, so QKRA uses about 57.4% as many counted MACs under that convention.

For batch 1, the asymptotic divergence is visible in the same accounting:

Context	QKRA major MACs	Transformer major MACs	QKRA/T
512	122.7M	213.9M	57.4%
1,024	245.4M	629.1M	39.0%
2,048	490.7M	2.064B	23.8%
4,096	981.5M	7.348B	13.4%
8,192	1.963B	27.582B	7.1%

A different but related quantity is logical score geometry. At context 512/batch 4, QKRA has $BNW = 196,608$ local score positions, while a literal four-head full-attention score tensor has $B \times 4 \times N \times N = 4,194,304$ positions. At context 8,192/batch 1 those counts are 786,432 versus 268,435,456. These counts are useful for understanding the mathematical relation geometry, but they are *not* claims that PyTorch physically materializes the Transformer’s full score tensor; fused attention kernels can avoid doing so.

6.4 Measured memory

The current implementation does not demonstrate lower peak process memory. QKRA is slightly higher in RSS in every isolated row above, including 336.2 MiB versus 320.9 MiB at context 8,192 inference. The correct current claim is therefore:

The ordinary QK relationship geometry and theoretical local workspace are much smaller, but the prototype has not yet converted this into lower measured whole-process RSS than PyTorch’s highly optimized fused Transformer implementation.

This distinction is important. Algorithmic storage and process-level memory are not the same measurement. Allocator behaviour, framework buffers, block temporaries, fused kernels and other model state can dominate the observed RSS at these model sizes.

6.5 Extra storage for the separate hop geometry

The separate hop path does add a small fixed parameter cost, but it does not require a second copy of the model or of the hidden-state sequence. In the hop-enabled final block the model stores four learned projection matrices,

$$W_{Q_0}, \quad W_{K_0}, \quad W_{Q_h}, \quad W_{K_h}. \quad (19)$$

The hop-specific pair therefore adds

$$\boxed{2d^2} \quad (20)$$

stored parameters. This is 4,608 parameters at $d = 48$ and 18,432 parameters at $d = 96$ (about 72 KiB in FP32). These hop parameters are already included in the matched totals reported in this note; the scaled QKRA and Transformer each contain exactly 271,296 parameters.

The hidden states H are shared by the ordinary and hop systems. During ordinary $k = 0$ operation, Q_h , K_h and A_h are not computed, so only their stored weights remain. When hopping is requested, extra temporary workspace is needed for Q_h , K_h and the current dense A_h . The dominant issue is therefore not the extra $2d^2$ weights but the present $N \times N$ hop field. The ordinary $A_0^{(W)}$ field is local and can be evaluated blockwise. The ordinary and hop fields should remain mathematically separate because they use different learned geometries, but their temporary workspace can be reused sequentially rather than kept live at the same time.

7 Why the linear ordinary path matters

The fixed-window change moves the default ordinary mode away from the sequence-length bottleneck of the earlier full-pair design. When $k = 0$, the architecture now has no dense relationship matrix at all. The local field uses at most W keys per query, and the separate hop projections remain dormant.

This creates a useful compute hierarchy:

ordinary prediction / reading
|

```

      v
fixed-window  $A_0^W$       -> cheap path; linear in  $N$  for fixed  $W$ 
      |
      +----- output

explicit deeper relation requested
      |
      v
full-history strict  $A_h$   -> expensive path; currently quadratic in  $N$ 
      |
      v
repeat  $A_h$   $k$  times, then local  $A_0^W$  read

```

The architecture therefore does not have to pay for full-history relational depth on every ordinary forward pass. Whether this becomes a large practical systems advantage depends on workload: an application that requests hops constantly would still spend substantial time in the quadratic path, while one that uses $k = 0$ for most ordinary processing and invokes deeper relations selectively would amortize the expensive component.

The fixed-window idea itself is not unique to QKRA. Sliding-window and sparse-attention Transformers can also achieve linear-in- N attention patterns. The distinctive aspect being explored here is the architectural separation between ordinary local reading and an independently learned, explicitly repeatable relationship operator. A future fair comparison should therefore include a separately trained local-window Transformer control, not only the full-attention control used to preserve continuity with the earlier paper.

8 What the experiments currently support

The evidence supports the following limited statements:

- The ordinary QK field can be restricted to $W = d$ while retaining the earlier small-benchmark language performance within the uncertainty of three seeds.
- The $W = 48$ change preserves the characteristic answer-only hop behaviour and the nearly perfect learned A_h transition field of the strict separate-hop architecture.
- A larger two-layer $d = 96$, $W = 96$ QK model with exactly 271,296 stored parameters finished ahead of a matched genuine four-head full-attention Transformer on loss and accuracy for all three tested seeds after 4,096 updates.
- The scaled benchmark uses actual local mathematics: the ordinary QK implementation does not build an $N \times N$ A_0 score matrix, and it numerically matches a dense-window reference to floating-point tolerance.
- In the measured CPU implementation, QKRA-W96 is faster in the reported ordinary-operation timing rows, with the observed advantage increasing at longer timing contexts.
- The ordinary path is $O(NWd)$ and therefore linear in sequence length for fixed W and d .
- The current explicit hop path remains quadratic because A_h is still discovered and propagated as a dense full-history relation field.

9 Limits of the current evidence

Important limits remain:

- The language corpus is still Vim documentation rather than a large standard language-model benchmark.
- Three seeds are useful evidence but remain a very small statistical sample. Intervals excluding zero in the scaled experiment should not be generalized into a universal superiority claim.
- The matched Transformer control is full-attention. A Transformer can also be given a fixed sliding window; a separately trained matched-window Transformer is a necessary follow-up control for isolating constant-factor architectural differences.
- The current A_h field remains $O(N^2)$ in sequence length when hops are requested. The architecture as a whole is not yet fully linear in hop mode.
- The long-context 1,024–8,192 measurements are speed probes. They do not establish language quality beyond the 512-token training context.
- Measured peak process RSS is not lower for the current QK prototype, despite the much smaller local relationship geometry.
- CPU timings should not be generalized to optimized GPU kernels or large production systems.
- The synthetic hop benchmark is a controlled compositional relation task, not evidence of general natural-language reasoning.
- The depth-33 terminal/readout boundary failure remains unresolved.
- Fixed-window ordinary reading does not by itself solve positional or context-length extrapolation.
- Autoregressive generation with caching has different per-token runtime behaviour from a full-sequence forward pass. The $O(N)$ versus $O(N^2)$ statements here refer to total ordinary relationship work over a length- N sequence for the stated implementations.

10 Next technical target: sparse hop discovery

The ordinary path is no longer the main asymptotic problem. The remaining target is A_h . In the controlled relation experiment, the learned strict hop field places essentially all top-1 decisions on the correct next relation, suggesting that its useful behaviour is much sparser than an arbitrary dense matrix.

One possible direction is to replace dense hop storage with one or a small number of learned relation pointers per position,

$$p(i) = \arg \max_{j < i} A_h(i, j), \quad (21)$$

or a sparse top- s relation set. If the correct distant relation could also be *discovered* without scanning every earlier key, repeated hopping could become sparse graph traversal rather than

dense matrix multiplication. That discovery problem is not solved in this note and should be treated as future work rather than part of the current linearity claim.

If successful, the longer-term target would be

$$\boxed{\text{linear local ordinary reading + sparse long-range relational propagation.}} \quad (22)$$

11 Publication, contact and licensing

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12 Conclusion

The fixed-window companion experiments materially change the scaling picture of the QK Relational Architecture’s ordinary path. Replacing full-pair A_0 with exact $A_0^{(W)}$ under the tested rule $W = d$ reduces ordinary relationship work from quadratic to linear in sequence length for a fixed model, without a detected quality penalty in the original $d = 48$ benchmark and with stronger-than-Transformer results in the larger $d = 96$, context-512 experiment.

The architecture now has a clear two-regime interpretation. Ordinary $k = 0$ operation uses only the fixed-window headless QK field and leaves the hop geometry dormant. Explicit deeper operation activates the separate strict full-history A_h field, which retains the reusable relational behaviour established in the earlier work but remains quadratic.

The strongest defensible interpretation is therefore not that the architecture is fully linear or that Transformers cannot be made linear. It is that the previously quadratic ordinary QK path has been replaced by a verified fixed-window form that scales linearly in sequence length, remains competitive in the tested language tasks, and produces real measured CPU speed savings at longer contexts. The next decisive problem is to obtain similarly sparse or subquadratic discovery and propagation for the optional A_h path without destroying its learned transition accuracy.